METRIC

MIL-PRF-32808
6 May 2026

PERFORMANCE SPECIFICATION

INCINERATOR, LARGE WITH SHRED-FEED SYSTEM, SHIPBOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers large shipboard incinerators that convert solid waste that is generated during the ship's normal service to ash by burning or high heat. A solid waste shred-feed system is required for use with this incinerator configuration to provide a metered feed. A capability for processing oily waste is not included in the scope of this specification. A capability for heat recovery from the flue gas is not included in the scope of this specification.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-5624	-	Turbine Fuel, Aviation, Grade JP-5 (NATO F-44)
MIL-PRF-17773	-	Switch, Bus Transfer, Electric Power, Automatic

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1399-300-1	-	Low Voltage Electric Power, Alternating Current
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

(Copies of these publications are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 4540

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:47Z
Check the source to verify that this is the current version before use.

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

593-6960891	-	Solid Waste Shredder Right Angle Speed Reducer Assembly and Details
593-6960928	-	Solid Waste Shredder Blast-Resistant Feed Hopper Details and Assembly
593-7556871	-	Mod 1 Solid Waste Shredder Chamber Assembly
593-7556873	-	Mod 1 Solid Waste Shredder Top Level Assembly
593-7649041	-	COTS Incinerator Day Tank Interface Control Drawing

(Copies of these publications are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

INTERNATIONAL MARITIME ORGANIZATION (IMO)

Resolution MEPC.244(66) - 2014 Standard Specification for Shipboard Incinerators

(Copies of this publication are available online at www.imo.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 General design. The incinerator system shall consist of the following:

a. Incinerator in accordance with Resolution MEPC.244(66) (see 3.2.1),
b. Shred-feed system (see 3.2.2),
c. Flue gas fan or other device for induced draft,
d. Flue gas damper (if incorporated),
e. Evacuation fan for the shred-feed system (if incorporated), and
f. Control panel.

3.2.1 Incinerator. The incinerator shall have an IMO type approval certificate in accordance with Resolution MEPC.244(66) (see 6.2).

3.2.2 Shred-feed system. The shred-feed system shall shred solid waste and provide a controlled feed to the combustion chamber. The shred-feed system shall be ventilated to prevent explosive gases from entering the combustion chamber. The shred-feed system may be entirely designed by the supplier or may include NAVSEA incinerator shredder assembly (593-7556873) as a component of the shred-feed system. The NAVSEA incinerator shredder assembly may be Government-furnished or supplier furnished and includes the feed hopper (593-6960928), cutter chamber (593-7556871), and cutter gearbox (593-6960891). Note, using the NAVSEA incinerator shredder assembly still requires the supplier to design a system to transfer shred to the combustion chamber and to ventilate explosive gases.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:47Z
Check the source to verify that this is the current version before use.

3.3 Dimensional.

3.3.1 Weight. The incinerator weight shall include the incinerator chamber, shred-feed system including shredder, flue gas exhaust fan or other induced draft device, flue gas damper, and control panels but shall not include the exhaust stack or fuel system. The weight of a large incinerator with shredder shall not exceed 9,072 kilograms (20,000 pounds).

3.3.2 Height. The deck footprint of the incinerator may be controlled if the incinerator will be required to fit inside an existing space (see 6.2). The height of the incinerator shall not exceed 2,159 millimeters (85 inches). The incinerator height shall include the incinerator chamber, shred-feed system including shredder, flue gas exhaust fan, flue gas damper, and control panels but shall not include the exhaust stack or fuel system.

3.3.3 Disassembled size. The incinerator unit shall be capable of disassembly into pieces (see 6.6) that will fit through a Navy standard door that is 660 by 1,676 millimeters (26 by 66 inches) with 203-millimeter (8-inch) radius corners and through a Navy standard hatch that is 762 by 1,524 millimeters (30 by 60 inches) with 203-millimeter (8-inch) radius corners.

3.4 Shipboard environment. The incinerator shall perform satisfactorily in the environmental conditions listed.

3.4.1 Ship motions. The incinerator shall be designed to operate under the ship motions specified in Resolution MEPC.244(66). This definition of ship motions shall take precedence over the requirements in the ship specification.

3.4.2 Electrical power. Electrical power shall be in accordance with MIL-STD-1399-300-1, type I.

3.4.3 Shock. The incinerator system shall be in accordance with the grade B, type A shock requirements of MIL-DTL-901.

3.4.4 Vibration. The incinerator system shall be in accordance with the type 1 vibration requirements of MIL-STD-167-1.

3.5 Interfaces.

3.5.1 Fuel. The incinerator shall be designed to burn JP-5 fuel in accordance with MIL-DTL-5624. The incinerator fuel will be supplied from an incinerator day tank (593-7649041).

3.5.2 Low pressure air treatment. The quality of ship's low pressure air (620 to 1,034 kilopascals [90 to 150 pounds per square inch gauge]) is uncontrolled with varying contaminants. If the incinerator requires low pressure air from the ship, at a controlled pressure or quality, the incinerator shall include air treatment capability and controls.

3.5.3 Combustion air supply. Air supply shall be drawn from the compartment that contains the incinerator.

3.5.4 Ventilation rate. The shred-feed system shall be vented with a minimum of 357 cubic meters per hour (210 cubic feet per minute) to prevent explosive gases from transferring to the furnace. If the incinerator requires control of the shred-feed system ventilation, the incinerator system shall include an explosion-proof ventilation fan capable of providing a minimum of 357 cubic meters per hour (210 cubic feet per minute). The incinerator shred-feed system shall include a means to connect to shipboard ductwork to vent explosive gases overboard. The ductwork will be provided by the Government during installation.

3.6 Operational. The incinerator shred-feed system shall be able to accommodate mixed waste consisting of cardboard, paper, plastic, and oily rags. The incinerator shall be able to process a waste stream consisting of the ratios shown in table I at the specified minimum processing rates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:47Z
Check the source to verify that this is the current version before use.

TABLE I. Operational rate.

Waste stream	Minimum processing rate (kilograms/hour [pounds/hour])
100% cardboard	136 (300)
75% cardboard:25% plastics	90.7 (200)
50% cardboard:50% oily rags	90.7 (200)

The incinerator draft and shred-feed system shall automatically adjust its operation to achieve a normal operating state with various feed changes including 100 percent plastic without operator intervention.

3.7 Safety and tolerance for misoperation.

3.7.1 Aerosol can with propellant. The incinerator shred-feed system shall limit combustible gases due to one aerosol can (or multiple cans) totaling 450 grams (15.9 ounces) of propellant that enters the combustion chamber to be no more than 25 percent of the lower explosive limit (LEL).

3.7.2 Explosion event mitigation. If an explosion were to occur, all explosive pressure shall be directed away from personnel and towards the overhead. In the event of an explosion, all parts of the incinerator and shred-feed system shall remain attached and shall not create the potential for personnel injury due to flying parts.

3.7.3 High temperature flue gas. Critical equipment exposed to flue gas, such as a flue gas fan and flue gas damper (if incorporated in the design), shall be capable of operating at temperatures of 800 °C (1,472 °F) for 20 minutes without damage.

3.7.4 Solid contaminants. The incinerator and shred-feed system shall tolerate metallic eating utensils, wire, and glass without damage and without affecting performance and safety. The quantity of these contaminants may be up to 7 percent by weight of the waste stream.

3.7.5 Automatic restarting of combustion chamber cooling. Power to the flue gas fan or other induced draft equipment will be routed separately from the power to the incinerator control panel. An alternate power source shall be provided for the flue gas fan/ejector via an Automatic Bus Transfer switch (ABT) in accordance with MIL-PRF-17773. The ABT will be Government-furnished material (see 6.7). If the induced draft equipment is not electrically operated, the means for ensuring continuity of operation shall be provided in a failure mode analysis and fire hazard analysis (see 6.5). On restoration of shipboard power, the exhaust fan or other induced draft equipment shall restart automatically if the combustion chamber temperature is above the maximum temperature set by the manufacturer.

3.7.6 Door interlocks. Door interlocks shall prevent the combustion chamber and ash removal doors from opening while burning is in progress or while the furnace temperature is above 150 °C (302 °F).

3.8 Ship interface information package. As specified (see 6.2), a technical documentation package shall be provided with the following information:

a. Foundation requirements for the incinerator and for the shred-feed system.

b. Dimensional, flow, and other requirements for the incinerator exhaust and shred-feed system exhaust.

c. Flow requirements for the ventilation air supply.

d. Flow requirements for the low pressure air supply.

e. Flow and quality requirements for cooling water (if incorporated).

f. Electrical power requirements for the incinerator and for the shred-feed system.

g. Data interface requirements such that key performance parameters and alarm states may be monitored remotely using the ship’s machinery control system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:47Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one unit. First article inspection shall consist of the examinations and tests specified in table II. Inspections will be witnessed and approved by the Technical Authority (see 6.2 and 6.8.1).

TABLE II. First article inspection.

Requirement	Requirement Paragraph	Test Paragraph
MEPC Compliance	3.2	4.2.1
General design	3.2	4.2.2
Weight	3.3.1	4.2.3
Height	3.3.2	4.2.3
Disassembled size	3.3.3	4.2.4
Ship motions	3.4.1	4.2.1
Electrical power compatibility	3.4.2	4.2.5.1
Shock	3.4.3	4.2.5.2
Vibration	3.4.4	4.2.5.3
Fuel compatibility	3.5.1	4.2.6
Low pressure air	3.5.2	4.2.7
Combustion air supply	3.5.3	4.2.8
Ventilation rate	3.5.4	4.2.9
Operational	3.6	4.2.10
Aerosol can with propellant	3.7.1	4.2.11.1
Explosion event mitigation	3.7.2	4.2.11.2
High temperature flue gas	3.7.3	4.2.11.3
Solid contaminants	3.7.4	4.2.11.4
Automatic restart	3.7.5	4.2.11.5
Door interlocks	3.7.6	4.2.11.6
Ship interface information	3.8	4.2.12

4.2.1 MEPC compliance. The incinerator shall be tested in accordance with the installation tests specified in Resolution MEPC.244(66). As specified (6.2), certification of compliance with Resolution MEPC.244(66) shall be provided.

4.2.2 General design. Vendor furnished information (see 6.2) shall be analyzed to determine that all necessary parts and subsystems are incorporated in the design, as specified in 3.2.

4.2.3 Weight and height. The incinerator weight and height shall be determined by analysis of vendor furnished information to verify compliance with 3.3.1 and 3.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:47Z
Check the source to verify that this is the current version before use.

4.2.4 Hatchability. The ability of the incinerator pieces to fit through a Navy standard door and through a Navy standard hatch shall be determined by analysis of vendor furnished information, to verify compliance with 3.3.3.

4.2.5 Shipboard environment.

4.2.5.1 Electrical power compatibility. Electrical power compatibility shall be determined by analysis of vendor furnished information, to verify compliance with 3.4.2.

4.2.5.2 Shock. The Incinerator system shall be tested in accordance with the grade B, type A shock requirements of MIL-DTL-901 to determine conformance to 3.4.3. The incinerator system shall be tested while operating under the following two conditions:

a. Normal Operation: with the control system energized, fuel lines under normal operating pressure, and all motors operating normally.

b. Cooling Operation: with the control system energized and flue gas fan (or other device for induced draft) operating.

4.2.5.3 Vibration. The equipment shall be subjected to the vibration tests in accordance with MIL-STD-167-1 to determine conformance to 3.4.4.

4.2.6 Fuel compatibility. Fuel compatibility shall be verified by manufacturer certification, to verify compliance with 3.5.1.

4.2.7 Low pressure air. If low pressure air is required, the air-operated elements shall be demonstrated.

4.2.8 Combustion air supply. The supply path for combustion air shall be determined by analysis of vendor furnished information, to verify compliance with 3.5.3.

4.2.9 Ventilation rate. The shred-feed system ventilation rate shall be measured using a vendor specified demonstration method to verify compliance with 3.5.4.

4.2.10 Incineration rates. The incineration rates for each waste stream mixture specified in table I shall be tested to verify compliance with 3.6. Each waste stream mixture in table I shall be tested for 120 minutes. Minimum processing rates less than those specified in table I shall constitute failure of this test. The ability to adjust automatically to waste stream variability shall be demonstrated to verify compliance with 3.6. The processing sequence for the waste stream variability test shall include 100 percent cardboard for 15 minutes, 100 percent plastic for 15 minutes, 100 percent cardboard for 15 minutes, 100 percent plastic for 15 minutes, followed by 15 minutes of no feeding, during which the temperature and combustion chamber pressure shall remain within system and MEPC limits.

4.2.11 Misoperation. The incinerator operation shall be demonstrated to tolerate the misoperation conditions specified in 3.7.1, 3.7.2, 3.7.3, 3.7.4, 3.7.5, and 3.7.6.

4.2.11.1 Aerosol can with propellant. LEL gas monitoring equipment with a response time of 1 second or less and with data logging shall be placed in the cold and unfired combustion chamber. An aerosol can or cans with propellant as required in 3.7.1 shall be fed into the shredder. The explosive gas level in the combustion chamber shall remain below the level as required in 3.7.1.

4.2.11.2 Explosive event mitigation. An ignition source shall be located immediately downstream of the shredder. An aerosol can or cans with propellant as required in 3.7.1 shall be fed into the shredder and the gas shall be ignited. The explosion vent path shall meet the requirements of 3.7.2 and all parts of the incinerator and shred-feed system shall remain attached without danger to the operator as required in 3.7.2.

4.2.11.3 High temperature flue gas. For equipment subject to 3.7.3, the manufacturer shall create the conditions and operate the critical equipment for the length of time required in 3.7.3. The equipment shall continue to operate until the equipment temperatures return to typical shutdown conditions. For systems using a flue gas fan, amplitudes of vibration shall be measured on the bearing housing in the direction of maximum amplitude. The equipment shall be inspected after the test. Any damage or exceeding the vibration thresholds specified in MIL-STD-167-1 after the equipment has cooled shall constitute failure of the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:47Z
Check the source to verify that this is the current version before use.

4.2.11.4 Solid contaminants. The incinerator and shred-feed system shall be tested with solid contaminants in the waste stream to verify the requirements of 3.7.4. The solid contaminants shall be metallic eating utensils, wire, and glass at a ratio up to 7 percent by weight of the waste stream. The incinerator and shred-feed system shall be inspected after the test and any damage shall constitute failure of the test.

4.2.11.5 Automatic restart of combustion chamber cooling. The exhaust fan or other induced draft equipment restart specified in 3.7.5 shall be tested. The incinerator shall be powered on and operated until the combustion chamber temperature reaches at least 150 °C (302 °F), actuating the door interlocks. After the temperature is reached, power to the incinerator shall be interrupted for 3 to 5 seconds after which power shall be restored. Failure of the exhaust fan or other induced draft equipment to restart after power restoration shall constitute failure of this test.

4.2.11.6 Door interlocks. The door interlock for the combustion chamber and ash removal shall be tested to verify compliance with 3.7.6. When the incinerator is burning, an attempt shall be made to open each door protected by a door interlock. When the incinerator is above the door interlock temperature, an attempt shall be made to open each door protected by a door interlock. Opening the door shall constitute failure of the test.

4.2.12 Ship interface information package. The adequacy and completeness of the interface information package required by 3.8 shall be verified by analysis of the package (see 6.2).

4.3 Conformance inspection. Each unit shall be tested in accordance with the factory and installation tests specified in Resolution MEPC.244(66). As specified (see 6.2), a certification of approval with Resolution MEPC.244(66) shall be provided.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Shipboard incinerators coved by this specification are intended for use in converting solid waste that is generated during the ship's normal service to ash by burning or high heat. Requirements of this specification are driven by the quantity and type of waste generated in the Navy environment and the prevention of damage to military personnel and assets. The incinerators covered by this specification are not intended to burn sludge, liquid waste, or hazardous materials. A shred-feed system is required for proper incinerator operation. The shredder components of the shred-feed system may be procured by invoking the requirements of 3.2 or may be procured separately.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. That first article inspection is required for all procurements (see 3.1).

c. Incinerator Certification of Approval in accordance with Resolution MEPC.244(66) (see 3.2, 4.2.1, and 4.3).

d. Special dimensional and arrangement requirements to suit shipboard needs, such as deck footprint dimensions, positioning of the control panel, or orientation of the flue gas fan or flue gas damper (see 3.3.2).

e. Shipboard interface information (see 3.8 and 4.2.12).

f. Additional first article testing provisions (see 4.2 and 6.4).

g. Requirements for submission of design information (see 4.2.2).

h. Packaging requirements (see 5.1 and 6.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:47Z
Check the source to verify that this is the current version before use.

i. Requirements for manufacturers offering an incinerator with novel features not addressed by this specification (see 6.5).

j. A copy of the NAVSEA approval letter that documents satisfactory first article inspection.

k. When on-site technical support from the manufacturer for installation and for commissioning tests is required.

l. When manufacturer-supplied training for the Navy operators is required.

m. When it is required for the manufacturer to offer for sale to the Navy the manufacturing, technical support, and logistics support information for the incinerator if the manufacturer elects to cease manufacture or support the incinerator model.

6.3 Installation integration. The incinerator and shredder specified in this document require ship support and ship features that are defined in the associated installation integration document.

6.4 Additional first article inspection provisions. Offerors must notify the Government in writing of the time and place of testing so the government may witness the testing. Technical witnesses should be obtained from the Technical Authority. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. FAR 52.209-3 applies (see 6.2).

6.5 Novel features. If a manufacturer offers an incinerator with novel features not addressed by the first article inspection requirements, a failure mode analysis and fire hazard analysis should be performed and used as the technical basis to tailor the first article inspection requirements. Technical concurrence with the necessity for further analysis, technical concurrence with the analyses, tailoring, and witnessing should be sought from the Technical Authority.

6.6 Additional hatchability provisions. The incinerator unit should be shipped in pieces (see 3.3.3).

6.7 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.2.

6.8 Definitions.

6.8.1 Technical authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05T may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.9 Subject term (key word) listing.

Shipboard waste

Shredder

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:47Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
DLA – CQ

Review activities:
Army – MI
Navy – YD
Airforce – 03
DLA – CC

Civil agency:
GSA – FAS

Preparing activity:
Navy – SH
(Project 4540-2026-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:47Z
Check the source to verify that this is the current version before use.